
**University of Puerto Rico
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Department of Electrical and Computer Engineering**



**ARTIFICIAL INTELLIGENCE
Technical Report: Group H
Agents Homework Assignment**

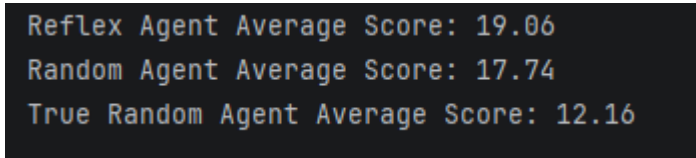
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ICOM5015
6 / March / 2026**

A. Exercise 2.11 discussion

The experimental results show clear differences in performance between the three agent programs tested in the vacuum world simulator. Each simulation was run with 100 trials in a defined spatial structure but with the primary location chosen at random. Three agents were created: a simple reflex agent that perceives when a space is dirty and cleans it and if not, moves to an adjacent space; a randomized reflex agent where movement is randomized yet still perceives the space for dirt; and a true randomized agent that randomly sucks and moves without perceiving if the space requires cleaning or not. The accuracy of each attempt was added and averaged, the discussion is based on this attempt although concurrent runs show similar results.



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Reflex Agent Average Score: 19.06  
Random Agent Average Score: 17.74  
True Random Agent Average Score: 12.16
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Figure 1: Results of Agent scores during simulation

In a run of the experiment, the simple reflex agent achieved the highest average score of 19.06. This result is expected because the reflex agent follows a deterministic rule that directly responds to the perception provided by the environment. When dirt is detected, the agent performs the suck action, which immediately improves the environment and increases the performance measure. When the square is clean, the agent moves to the other location, ensuring that both squares are periodically checked. Because the environment consists of only two locations and the percept fully describes the relevant state of the current square, this simple rule results in efficient behavior and consistently high performance.

The randomized reflex agent obtained a slightly lower average score of 17.74. This agent still cleans the square whenever dirt is detected but selects movement actions randomly regardless of whether the square is clean or not. While this introduces some variability that can help explore the environment, it also leads to unnecessary actions like repeated movements or executing NoOp actions. Since the performance measure penalizes actions and rewards maintaining clean squares, these extra movements slightly reduce the agent's overall score compared to the simple reflex agent's policy.

The true random agent performed significantly worse, with an average score of 12.16. This agent acts entirely random, choosing every action without structure and without perception of the environment. As a result, it may fail to clean a square even when dirt is present or clean an already cleared square, creating many false positives and negatives. Because it does not make decisions based on the environment state, it is unable to maintain cleanliness efficiently, which leads to substantially lower cumulative performance compared to the other agents who perceive their environment to be.

Overall, the results demonstrate the importance of using percept information when designing agent programs. Agents that respond to the environment in a structured way achieve higher performance than agents that act randomly. Even simple decision rules based on the current perception can significantly improve the effectiveness of an agent in maintaining a clean environment. This is not to say that utilizing randomness as a means to explore an environment is not unviable for other experimentations, but for more controlled or known environments, perception is crucial to the efficiency of the simulated Agents.

B. Exercise 2.14 discussion

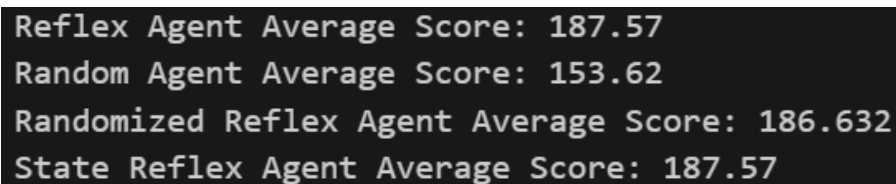
1) Can a reflex agent be perfectly rational?

Here, the vacuum world has unknown size, boundaries, obstacles, and dirt configuration. A simple reflex agent chooses actions based only on the current percept and has no internal state or memory. However, acting rationally in this environment requires remembering previously visited locations and explored areas in order to systematically explore the environment and avoid redundant movements. Because the optimal action depends on information about past percepts and previously visited states, the correct action cannot be determined from the current percept alone. Therefore, a simple reflex agent cannot achieve perfect rationality in this environment.

2) Can a simple reflex agent with a randomized agent function outperform a simple reflex agent?

The comparison between a simple reflex agent and a simple reflex agent with a randomized agent function illustrates the importance of structured decision-making in autonomous systems. A simple reflex agent follows a fixed set of condition–action rules that guide its behavior based on the current percept. For example, in this case it may clean a location when dirt is detected and otherwise move in a consistent direction according to a predefined strategy. This rule-based approach allows the agent to behave in a purposeful and predictable manner.

In contrast, a randomized agent selects actions without a consistent strategy, choosing among possible actions at random. Although this agent may occasionally perform useful actions, such as cleaning a dirty location, its behavior lacks direction and coordination. As a result, it is more likely to perform unnecessary movements or revisit locations inefficiently.



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Reflex Agent Average Score: 187.57
Random Agent Average Score: 153.62
Randomized Reflex Agent Average Score: 186.632
State Reflex Agent Average Score: 187.57
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Figure 2: Results of agent scores during simulation

The key difference between the two approaches lies in the quality of the decision-making process rather than the computational resources required to execute it. Both agents can operate with similar computational efficiency, but the reflex agent benefits from a structured policy that guides its actions toward achieving the task more effectively.

Overall, the simple reflex agent generally outperforms the randomized agent in this environment because its rule-based behavior aligns with the goals of the environment, while the randomized agent's undirected action selection leads to inefficient performance.

3) Can you design an environment in which your randomized agent will perform poorly?

An environment in which a randomized agent performs poorly is one with a maze-like structure. The fundamental challenge for a randomized agent in such an environment arises from its lack of directed action and memory. Because the agent selects actions randomly and does not retain information about previous states, it cannot use the structure of the environment to guide its behavior.

In a maze, movement is constrained by walls and narrow pathways. Random action selection therefore leads to a high probability of revisiting previously explored locations or repeatedly attempting actions that are blocked by walls. As a result, the agent may spend a large amount of time wandering without making meaningful progress toward unexplored areas or any potential goal within the environment.

The structure of a maze amplifies the weaknesses of a random strategy. While a more deliberate agent could learn to avoid dead ends, remember explored paths, or systematically search the environment, a randomized agent continues to move arbitrarily. This lack of guidance leads to inefficient exploration and wasted actions, making maze-like environments particularly challenging for agents that rely solely on random decision-making.

4) Can a reflex agent with state outperform a simple reflex agent? Can you design a rational agent of this type?

A reflex agent with state can outperform a simple reflex agent because it maintains an internal representation of aspects of the environment that are not directly observable from the current percept. While a simple reflex agent selects actions solely based on the current percept, a reflex agent with state can incorporate information from previous percepts and actions to guide its decisions.

The use of internal state allows the agent to remember previously visited locations, track which areas have already been cleaned, or maintain knowledge about the environment. This additional information enables the agent to avoid redundant actions and make more informed decisions, which can improve performance in environments where the optimal action cannot be determined from the current percept alone.

Although both agents may perform similarly in very simple environments, such as small and fully observable vacuum worlds, the reflex agent with state is generally better suited for more complex or partially observable environments. In such cases, the ability to maintain and update an internal model of the environment allows the agent to behave more efficiently and achieve higher performance.

A rational agent of this type can be designed by defining an internal state that records relevant information about the environment, such as which locations have been visited or cleaned. The agent updates this internal state after each percept and uses it, together with the current percept, to select an action that maximizes its expected performance according to the task's performance measure.

References:

[1] S. J. Russell and P. Norvig, Artificial Intelligence: A Modern Approach. Upper Saddle River: Prentice-Hall, 2010.

[2] GeeksforGeeks, "Simple reflex agents in AI," GeeksforGeeks, <https://www.geeksforgeeks.org/simple-reflex-agents-in-ai/> (accessed Mar. 5, 2026)